Weekly Progress Report

Refactoring Detection Project - Week 4 progress update for the Monday research supervisor meeting

1. Overview

During Week 4, the team moved from a finished IntelliJ-style UI design into implementation planning and early UI work. The main focus was confirming that all members could run the AntiCopyPaster tool, reviewing the current UI Panel with runIDE, deciding how to handle multiple clone cases, and dividing implementation responsibilities across the team. The report also documents the GitHub workflow the team agreed to use for separate branches, pull requests, review, and merging into the ui_panel branch.

The Week 4 progress is based on the Week 4 meeting notes, the Week 4 slide deck, and the team implementation planning discussion.

2. Weekly Progress Summary

Area	Progress Made
Tool running	All team members were able to run the tool. The main remaining issue was API token limits and provider usage limits.
UI design	The UI design was finished using the IntelliJ Figma Kit so the prototype matches the IntelliJ IDE style more closely.
Trust score	The Confidence/Trust Score direction was updated by Dhir and kept focused on workflow evidence such as clone detection, usefulness, compilation, and testing.
UI Panel review	The team planned to use runIDE and Dr. AlOmar's demo video to compare the current UI Panel with the Figma design.
Implementation planning	UI implementation tasks were divided across the team, and each person was assigned a clear section to work on.
GitHub workflow	The team agreed to work on separate branches, open pull requests into ui_panel, review changes, fix conflicts, and then merge accepted work.

3. Professor Meeting Context and Completed Tasks

In the fourth meeting on 06/08/2026, the team presented the Week 3 slides and reviewed the progress made since the previous meeting. The team confirmed that all members were able to run the tool, with API token limits being the main issue left. The UI design using the IntelliJ Figma Kit was finished, and the updated Confidence/Trust Score work was reviewed.

The main tasks from Dr. AlOmar were to add UI support for given cases, especially more than one clone, use runIDE to inspect the current UI Panel, review the demo video shared on Discord, and divide UI implementation tasks.

Meeting Item	Week 4 Update
Week 3 slides	The team presented the third week's progress slides, including tool testing, UI redesign, log review, and AI Trust Score direction.
Tool running	All members were able to run the AntiCopyPaster tool. The main remaining issue involved API token limits.
UI design	The UI design was finished with the IntelliJ Figma Kit and is ready to be compared with the current UI Panel.
Trust score	Dhir updated the Confidence/Trust Score direction, keeping it connected to workflow validation evidence.
Implementation direction	The team moved toward implementing the UI Panel and dividing the work into manageable sections.

4. Team Meeting: UI Implementation Planning

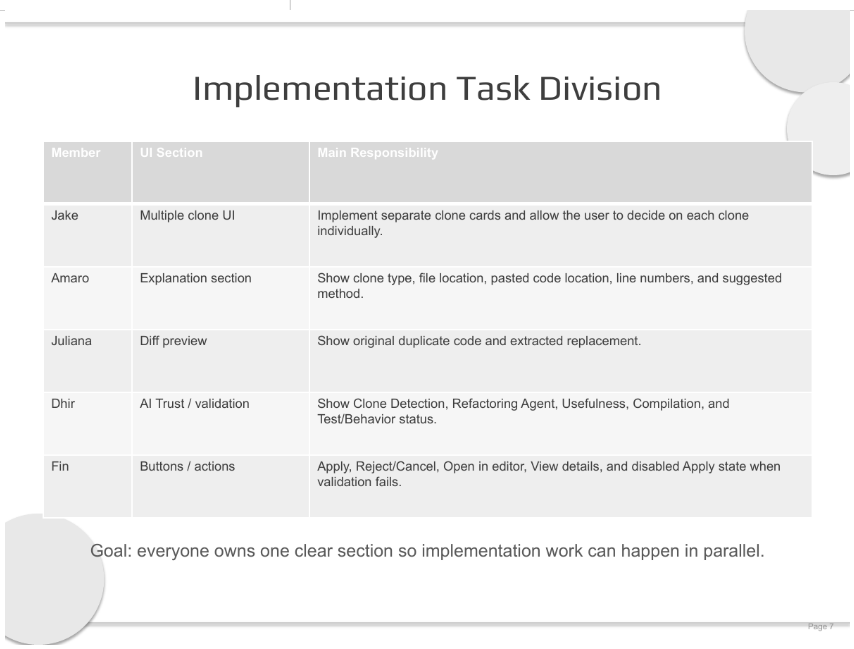
During the Wednesday team meeting, the team focused on how to start implementing the UI Panel and how to divide the work clearly. The main goals were to add support for cases shown by the tool and logs, use runIDE to understand the current UI Panel, review Dr. AlOmar's demo video, and assign each person a clear UI section.

Decision	Details
Starting point	Use the current UI Panel branch as the implementation base.
runIDE review	Everyone should run the project with runIDE to understand the current UI. If possible, one person should send screenshots or a screen recording on Discord.
Design comparison	Compare the current UI Panel with the finished Figma design before making larger implementation changes.
Important UI sections	Keep the explanation section, diff preview, AI Trust/validation section, action buttons, and details/provenance.

5. Implementation Task Division

The team divided the UI implementation work so each member owns one section. This allows everyone to work in parallel while still keeping the project organized.

Team Member	Assigned UI Section	Main Responsibility
Jake Stinton	Multiple clone UI	Implement separate clone cards and allow the user to decide on each clone individually.
Amaro da Luz	Explanation section	Show clone type, file location, pasted code location, line numbers, and suggested method.
Juliana	Diff preview	Show original duplicate code and extracted replacement.
Dhir Rana	AI Trust / validation	Show Clone Detection, Refactoring Agent, Usefulness, Compilation, and Test / Behavior status.
Fin Naughton	Buttons / actions	Apply, Reject/Cancel, Open in editor, View details, and disabled Apply state when validation fails.


Implementation Task Division

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Amaro	Explanation section	Show clone type, file location, pasted code location, line numbers, and suggested method.
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Dhir	AI Trust / validation	Show Clone Detection, Refactoring Agent, Usefulness, Compilation, and Test/Behavior status.
Fin	Buttons / actions	Apply, Reject/Cancel, Open in editor, View details, and disabled Apply state when validation fails.

Goal: everyone owns one clear section so implementation work can happen in parallel.

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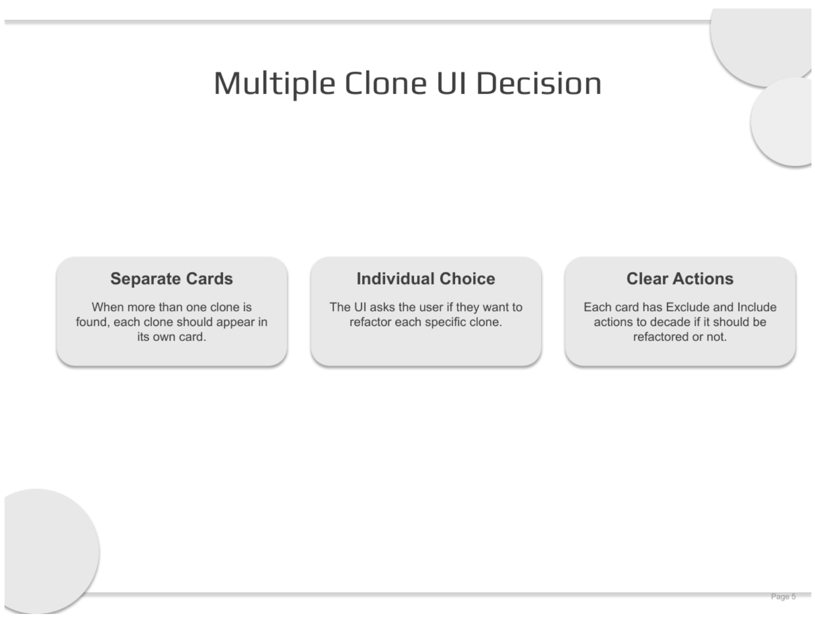
6. Multiple Clone UI Decision

The team decided that when the tool finds more than one clone, the UI should show each clone in a separate card instead of combining all results into one section. Each card should represent one specific clone or refactoring suggestion. This gives the developer more control and makes the UI clearer when multiple duplicate-code cases are detected.

UI Choice	Reason
Separate clone cards	Each clone appears in its own card so the user can review one suggestion at a time.
Individual decision	The UI asks the user individually whether they want to refactor each clone.
Clear card details	Each card should show clone location, line numbers, short explanation, suggested Extract Method refactoring, AI Trust / validation status, and action buttons.
User control	The user can apply, reject, or view details for each clone instead of being forced to refactor all clones at once.

Multiple clone flow: More than one clone found -> show separate clone cards -> user reviews each card individually -> user chooses Apply / Reject / View Details for each specific clone.



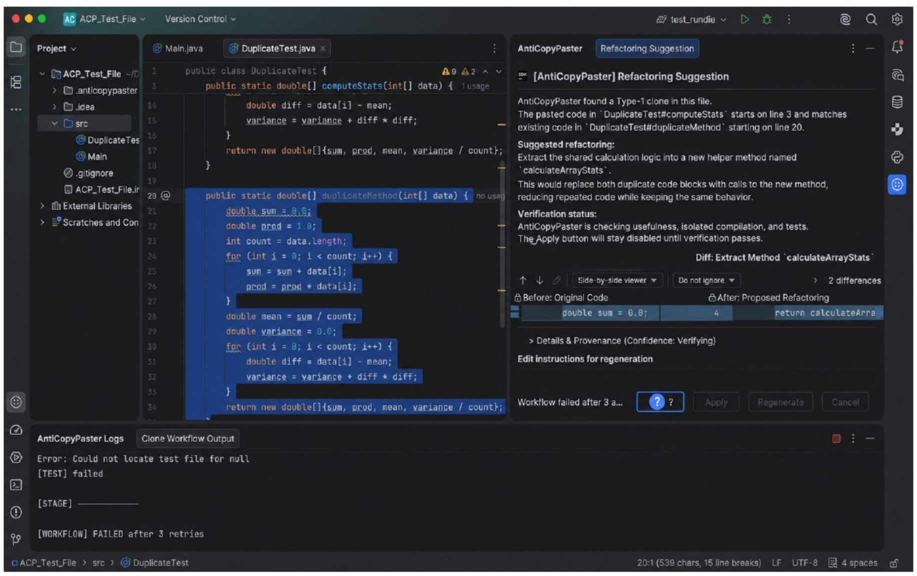
7. Explanation Section Updates

Amaro was assigned to improve the explanation section. The goal is to make the refactoring suggestion easier to understand before the developer decides whether to inspect, apply, or reject it. The updated text should clearly show the clone type, file locations, pasted-code location, line numbers, method names, and suggested Extract Method refactoring.

Goal	Why It Matters
Readable explanation	The developer should quickly understand what AntiCopyPaster found without reading raw logs.
Clear clone metadata	Clone type, existing location, pasted location, methods, and line numbers should be easy to scan.
Simple refactoring summary	The UI should explain that the repeated logic will be extracted into a helper method.
Decision support	The section should help the user decide whether to inspect, apply, or reject the refactoring.

The improved wording describes a Type-1 clone, identifies the pasted code in **DuplicateTest#computeStats**, identifies the existing clone in **DuplicateTest#duplicateMethod**, and explains the suggested helper method **calculateArrayStats**.

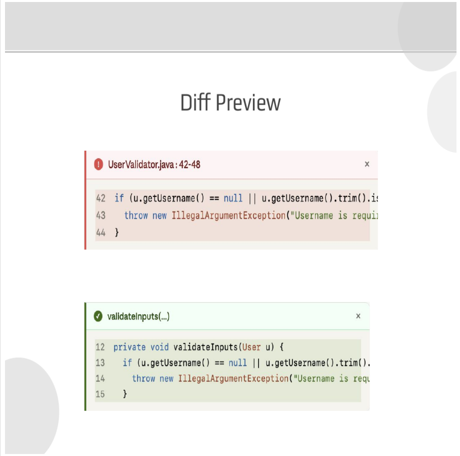
Updated

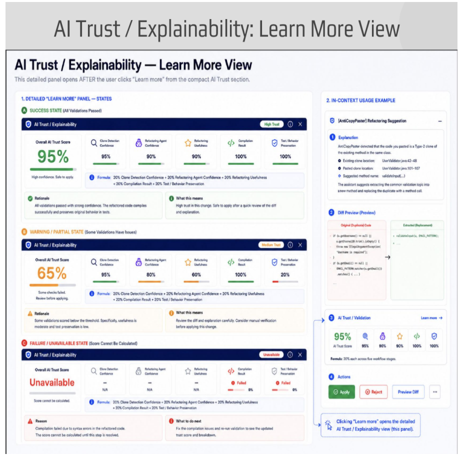


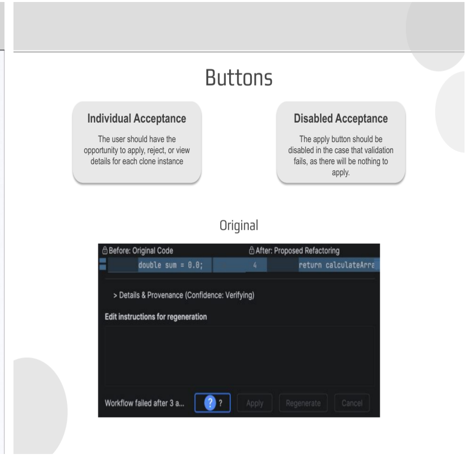
8. Diff Preview and Validation UI Direction

The diff preview should show the original duplicate code beside the proposed replacement so the developer can understand exactly what will change. The AI Trust / validation section should continue showing workflow evidence such as Clone Detection, Refactoring Agent, Usefulness, Compilation, and Test / Behavior status.

UI Section	Week 4 Direction
Diff preview	Show original duplicate code and extracted replacement in a clear side-by-side style.
AI Trust / validation	Show each validation component separately instead of hiding the result inside one score.
Buttons / actions	Apply, Reject/Cancel, Open in editor, View details, and disabled Apply when validation fails.
Details / provenance	Keep technical details available for users who want to inspect the suggestion more deeply.


Diff Preview
UserValidator.java : 42-48
42 if (u.getUsername() == null || u.getUsername().trim().is
43 throw new IllegalArgumentException("Username is requi
44 }
validateInputs(...)
12 private void validateInputs(User u) {
13 if (u.getUsername() == null || u.getUsername().trim().
14 throw new IllegalArgumentException("Username is requ
15 }

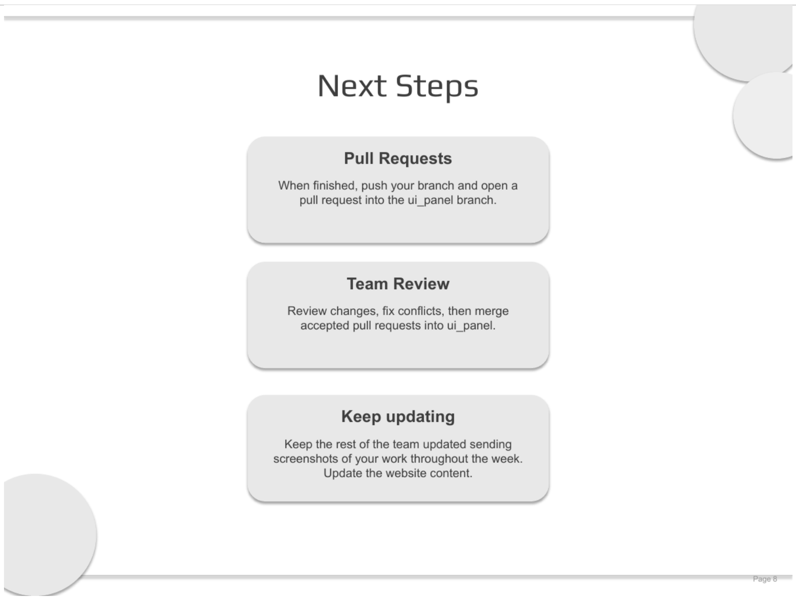

AI Trust / Explainability: Learn More View
AI Trust / Explainability — Learn More View
95%
65%
Unavailable


Buttons
Individual Acceptance
The user should have the opportunity to apply, reject, or view details for each clone instance
Disabled Acceptance
The apply button should be disabled in the case that validation fails, as there will be nothing to apply.
Original
Before: Original Code
After: Proposed Refactoring
double sum = 0.0;
return calculateArra
> Details & Provenance (Confidence: Verifying)
Edit instructions for regeneration
Workflow failed after 3 a...
Apply
Regenerate
Cancel

9. GitHub Workflow and Collaboration Process

The team decided to use separate branches and pull requests so that UI work can be reviewed before it is merged into the shared UI Panel branch. This helps avoid overwriting each other's work and makes it easier to check conflicts.

Step	Team Workflow
1. Separate branch	Each person creates or uses their own branch for their assigned UI section.
2. Push to GitHub	When a section is ready, the member pushes their branch to GitHub.
3. Pull request	The member opens a pull request into the ui_panel branch.
4. Team review	The team reviews the changes, checks for conflicts, and asks for fixes if needed.
5. Merge	Accepted pull requests are merged into ui_panel. After that, everyone pulls the latest ui_panel branch locally.



10. Next Steps

- Each member should confirm their assigned section and continue working on their own branch.
- Use runIDE to inspect the current UI Panel and share screenshots or a screen recording on Discord when possible.
- Compare the current UI Panel with the finished Figma design before making larger changes.
- Open pull requests into the ui_panel branch when sections are ready for review.
- Review team pull requests, fix conflicts, and merge accepted work into ui_panel.
- Continue updating the website with weekly files, including slides, notes, report, collaborators, and implementation screenshots.

11. Conclusion

Overall, Week 4 moved the project from finished UI design into implementation planning and early implementation work. The team confirmed that everyone could run the tool, completed the IntelliJ-style Figma design, refined the trust score direction, decided how multiple clone cases should be shown, and divided the UI work into clear sections. The next focus is to implement these sections on separate branches, review them through pull requests, and merge accepted work into the ui_panel branch while continuing to update the progress website.